

YIMENG WU

Hong Kong SAR

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Summary

Researcher with over 5 years of experience in large language model (LLM) pretraining. Proven expertise in scaling billion-parameter models, including Mixture-of-Experts (MoE) architectures and multilingual LLMs (e.g., Arabic JABER, SABER, AraMUS). Published multiple first-author papers in top-tier NLP conferences (EMNLP, AAAI, ACL). Strong background in model analysis, training stability, and performance tuning on advanced hardware (e.g., Huawei NPUs).

Education

- **McGill University** Montreal, QC, Canada
M.Eng. in Electrical and Computer Engineering, GPA: 3.77/4.0 Sept. 2017 - May 2019
- **Tianjin University** Tianjin, China
B.Eng. in Biomedical Engineering, GPA: 84/100 Sept. 2013 - July 2017

Work Experience

- **Huawei Noah's Ark Lab Hong Kong** Hong Kong
Researcher: Work on MoE pretraining May 2023 - present
- **Huawei Noah's Ark Lab Canada** Toronto, ON
Associate Researcher: Work on Arabic LLM projects June 2021 - May 2023
- **Huawei Noah's Ark Lab Canada** Montreal, QC
Support Researcher: Work on knowledge distillation and translation projects Dec. 2019 - June 2021

Projects

- **Pangu Large Language Model (LLM) Pretraining** May 2023 - present
 - Manage the pretraining of the 700B Mixture-of-Experts (**MoE**) model, track and analyze the model configuration including parameters, gradients, load balancing, and routing results.
 - Incorporate advanced algorithms into the MoE model, including the dynamic sparse routing, the design of auxiliary loss, and the sparsity ratio of models.
 - Conduct extensive analysis on dense and MoE models, ranging from million parameters to **billion** parameters, to explore the scaling laws on Huawei Ascend cards.
- **Arabic LLM Pretraining** June 2021 - Apr. 2023
 - Pretrain a series of Arabic LLMs on 50B textual data for commercial use, including JABER (BERT-base), SABER (BERT-large), and AraMUS (**T5-11B**).
 - Solve the **training instability** issues of AraMUS by enhancing the model architecture, calibrating the mixed precision training strategies, normalizing the key vector in the attention module, etc.
 - AraMUS gets **Rank 1** on a public Arabic leaderboard (ALUE), which contains eight NLP/NLU tasks and outperforms the second one by 4% on the average score. It also shows impressive few-shot learning abilities compared with the best existing Arabic LLMs.
- **Robust Machine Translation** Sept. 2020 - Dec. 2020
 - Train separate $En \rightleftharpoons Zh$ on-cloud and on-device transformer models that are robust to OCR noise.
 - Analyze OCR errors inside the training data and extract noise patterns. The training set is augmented with 80 million noise samples, which are generated based on the extracted patterns.
 - Improve the BLEU score averagely by 4 and 0.6, respectively, on internal $En \rightarrow Zh$ and $Zh \rightarrow En$ OCR test sets, without degradation regarding inference speed and performance on regular test sets.
- **Intermediate Layer Knowledge Distillation** Dec. 2019 - June 2021
 - Propose a **combinatorial-layer** knowledge distillation (KD) method to distill from intermediate layers of neural networks and final predictions. Furthermore, it is the first attempt that the combination is leveraged for KD approaches. (accepted by EMNLP 2020)

- Propose an automatic layer combination technique in intermediate KD which relies on **attention** mechanisms. (accepted by AAAI 2020)
- Propose a general intermediate KD method, which matches the intermediate layers of the teacher and the student in the **output space** via the attention-based layer projection, instead of matching in the hidden spaces. (accepted by EMNLP 2021)

Publications

- **Yimeng Wu**, Mehdi Rezagholizadeh, Abbas Ghaddar, Md Akmal Haidar, and Ali Ghodsi. “Universal-KD: Attention-based Output-Grounded Intermediate Layer Knowledge Distillation.” In Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (**EMNLP** 2021, **Oral**, acceptance rate: 25.6%)
- **Yimeng Wu***, Peyman Passban*, Mehdi Rezagholizadeh, and Qun Liu. “Why Skip If You Can Combine: A Simple Knowledge Distillation Technique for Intermediate Layers.” In Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (**EMNLP** 2020, short, acceptance rate: 16.7%)
- Abbas Ghaddar*, **Yimeng Wu***, Sunyam Bagga, Ahmad Rashid, Khalil Bibi, Mehdi Rezagholizadeh, Chao Xing et al. “Revisiting Pre-trained Language Models and their Evaluation for Arabic Natural Language Understanding.” (**EMNLP** 2022, acceptance rate: 22.1%)
- Pangu Team. “**Pangu Ultra MoE**: How to Train Your Big MoE on Ascend NPUs.” preprint (2025)
- Marzieh Tahaei, Aref Jafari, Ahmad Rashid, David Alfonso-Hermelo, Khalil Bibi, **Yimeng Wu**, Ali Ghodsi et al. “Efficient Citer: Tuning Large Language Models for Enhanced Answer Quality and Verification.” (**NAACL** findings 2024, acceptance rate: 12.5%)
- Asaad Alghamdi, Xinyu Duan, Wei Jiang, Zhenhai Wang, **Yimeng Wu**, Qingrong Xia, Zhefeng Wang et al. “AraMUS: Pushing the Limits of Data and Model Scale for Arabic Natural Language Processing.” (**ACL** findings 2023, acceptance rate: 19.1%)
- Passban Peyman, **Yimeng Wu**, Mehdi Rezagholizadeh, and Qun Liu. “ALP-KD: Attention-Based Layer Projection for Knowledge Distillation.” In Proceedings of the AAAI Conference on Artificial Intelligence (**AAAI** 2021, acceptance rate: 21%)
- Xian Shuai, Yiding Wang, **Yimeng Wu**, Xin Jiang, and Xiaozhe Ren. “Scaling Law for Language Models Training Considering Batch Size.” preprint (2024)
- Abbas Ghaddar, **Yimeng Wu**, Ahmad Rashid, Khalil Bibi, Mehdi Rezagholizadeh, Chao Xing, Yasheng Wang et al. “JABER: Junior Arabic BERT.” preprint (2022)

Awards

- HKRC Star Individual Award, 2024
- Huawei Noah’s Ark Lab Commercial Contribution Award, 2022

Professional Services

- **Conference Reviewers**
 - Annual Meeting of the Association for Computational Linguistics (ACL) 2022-2025
 - Conference on Language Modeling (COLM) 2024-2025
 - Empirical Methods in Natural Language Processing (EMNLP) 2022-2024
 - Natural Language Processing and Chinese Computing (NLCC) 2024
 - Efficient Natural Language and Speech Processing (ENLSP NeurIPS Workshop) 2021-2023

Skills

- Programming Language: Python, Matlab, Java
- ML Frameworks: Pytorch, Tensorflow, Megatron, Deepspeed, HuggingFace, Mindspore
- ML Areas: Natural language processing (NLP), Deep Learning, Medical Image & Signal Processing